Literature Review

Active Learning Platform for Selecting Data for Human Labeling

DEPI - AWS Machine Learning Engineer Track | Graduation Project

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1. Introduction

Modern deep learning models depend on large labeled datasets, but labels are slow and expensive to obtain, especially when they must come from human annotators. Active learning (AL) addresses this by letting the learning algorithm choose which unlabeled examples are worth labeling, so that a given accuracy is reached with fewer labels [1]. This review summarizes the main ideas, query strategies and open problems in (deep) active learning, and the tooling for human labeling on AWS, in order to justify the design choices of our platform.

2. Background: The Active Learning Setting

In the pool-based scenario, which fits our project, a learner starts from a small labeled set and a large pool of unlabeled data. It repeatedly selects informative examples from the pool, asks an oracle (a human annotator) to label them, and retrains [1]. Two other scenarios are described in the literature: membership query synthesis (the learner generates its own queries) and stream-based selective sampling (examples arrive one at a time and the learner decides whether to query) [1]. Early foundational work includes uncertainty sampling for text classification [2] and the query-by-committee framework, where disagreement among a committee of models identifies informative examples [3]. Settles' survey [1] remains the standard reference, and Ren et al. [5] provide a comprehensive survey of deep active learning, covering query strategies, batch selection and applications.

3. Query Strategies

3.1 Uncertainty-based sampling

The learner queries the examples it is least sure about [1], [2]. For a classifier with predicted class probabilities, three common measures are: **least confidence** (one minus the highest class probability), **margin** (the gap between the two most likely classes, where a small gap means high uncertainty) and **entropy**, the Shannon entropy of the predicted distribution [4]. These are cheap to compute because they need only one forward pass over the pool, which makes them the natural first choice for our platform.

3.2 Committee, ensemble and Bayesian methods

Query-by-committee [3] selects examples where several models disagree. For deep networks, Gal and Ghahramani showed that dropout at inference time approximates Bayesian uncertainty [6], and Gal et al. used it with acquisition functions such as BALD and max-entropy for image classification [7]. Beluch et al. found that ensembles of networks gave better uncertainty estimates and better active learning results than Monte Carlo dropout on image classification tasks [8], at the cost of training several models.

3.3 Diversity-based and hybrid methods

Selecting the top-N most uncertain examples can return near-duplicates, which wastes labels. Core-set selection [9] frames active learning as choosing a subset that represents the whole pool in feature space. BADGE [10] combines uncertainty and diversity by sampling from gradient embeddings, and BatchBALD [11] extends BALD to score a whole batch jointly. These methods can be more label-efficient in some settings but are more complex and costlier to run.

3.4 Learning-based methods

Yoo and Kweon [12] train a small module that predicts the loss of unlabeled examples and query those with the highest predicted loss, avoiding explicit uncertainty estimates.

Strategy	Main idea	Strengths	Limitations	Ref.
Least confidence / margin / entropy	Query examples with the most uncertain predictions	Simple, one forward pass, easy to implement	Overconfident networks; redundant batches	[1], [2], [4]
MC dropout / BALD	Approximate Bayesian uncertainty	Principled uncertainty, single model	Multiple stochastic passes	[6], [7]
Deep ensembles	Disagreement between several networks	Strong uncertainty estimates	Training cost multiplied	[8]
Core-set	Pick a representative subset in feature space	Promotes diversity	Distance computation on large pools	[9]
BADGE / BatchBALD	Combine uncertainty and diversity per batch	Good batch quality	More complex, costlier	[10], [11]
Learning loss	Predict the loss of unlabeled examples	No explicit uncertainty needed	Extra module to train	[12]

4. Challenges in Deep Active Learning

- **Overconfident predictions.** Modern neural networks are often poorly calibrated, so softmax-based uncertainty can be misleading; calibration methods such as temperature scaling help [13].
- **Batch redundancy.** Labeling is done in rounds, so selecting a batch of similar uncertain examples wastes the labeling budget; diversity-aware methods address this [9], [10], [11].
- **Cold start.** With very few initial labels, uncertainty estimates are unreliable. Hacohen et al. show that with a low budget, selecting typical (representative) examples works better than selecting uncertain ones, while uncertainty-based selection works better with a higher budget [14].
- **Reproducibility.** Reported gains depend on training details; Munjal et al. show that results can change substantially with regularization and hyperparameters, and that random sampling is a strong baseline when well tuned [15]. This motivates comparing every strategy against a random baseline under identical settings.

5. Human Labeling Tooling on AWS

Amazon SageMaker Ground Truth provides labeling workflows with human workforces, and an automated data labeling feature that uses active learning to decide which objects go to humans and which are labeled automatically [18]. According to the AWS documentation, automated labeling is intended for large datasets (a minimum of 1,250 objects, with at least 5,000 recommended) and incurs SageMaker training and inference costs [18].

Important availability constraint. The AWS documentation states that Ground Truth is closed to new customers effective 30 July 2026, that existing customers can continue to use it, and that no new features are planned [18]. A newly created AWS account may therefore be unable to use Ground Truth, so the labeling component of our design must not depend on it. Alternatives to evaluate in the design phase include a self-hosted open-source labeling tool, or a lightweight custom labeling web application on AWS (for example S3, API Gateway, Lambda and DynamoDB with a simple front end) used by team members as annotators. In all cases the active learning selection logic remains our own implementation on SageMaker and Lambda.

6. Datasets and Models

CIFAR-10 [16] is a widely used image classification benchmark with 60,000 32x32 color images in 10 classes, and is a common choice in active learning experiments [7], [9], [10], [15]. ResNet architectures [17], such as ResNet-18, are standard backbones in these studies. We will use CIFAR-10 with a ResNet-18 style model as the initial benchmark because it allows fast, repeatable experiments on a small AWS budget.

7. Gap and Positioning of Our Project

Most of the literature evaluates strategies offline, with all labels already available and hidden from the learner. Our project contributes a practical, cloud-based implementation: an automated loop on AWS (train, score, select, label, retrain) with a human-in-the-loop labeling step, a comparison of several query strategies against a random baseline under the same budget, and tracking of accuracy, labels used and AWS cost per iteration. Given the constraints above, we will start with uncertainty-based strategies (least confidence, margin, entropy) and a random baseline, and treat MC dropout, ensembles or diversity-aware selection as optional extensions if time allows.

8. Conclusion

The literature shows that active learning can reduce labeling effort, that simple uncertainty sampling is a strong and practical starting point, and that careful baselines, cold-start handling and batch diversity are essential for trustworthy results. Together with the limited availability of SageMaker Ground Truth for new accounts, these findings shape our design: an uncertainty-sampling loop with a random baseline, built on SageMaker and Lambda, with a labeling step that does not depend on Ground Truth.

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